

A04 Reflection Journal – Wizards vs Aliens

Team: DataProxy

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Working on the A04 project helped our group understand **weights, bias, and activation functions** in a way that felt much easier than just memorizing definitions. Instead of explaining everything with complicated formulas, we turned the topic into a simple game called **Wizards vs Aliens**. The main idea was that the wizard had to decide whether to **ATTACK or WAIT**, and the AI used simple clues to make that choice. That made the topic feel more realistic because we could actually see how each part of the AI decision worked instead of only reading about it. One of the first things we focused on was the idea of **clues**. In the presentation, the AI looks at things like the wizard's **health, armor, and how close the boss is**. We used those clues because they were easy to understand and because they showed that AI has to look at information before making a decision. This helped us understand that an AI system does not just randomly choose an answer. It looks at the information it is given, then uses that information to decide what matters most. In our game, health and armor were simple examples of the information the AI could use.

The **weights** part became clearer when we turned importance into stars. In our example, **health had two stars and armor had one star**. That meant health was more important to the AI. We explained this as “more stars = more important,” which made the meaning of weights much easier to understand. Instead of thinking of weights as just numbers inside a neural network, we could see them as a way for the AI to decide which clue deserves more attention. The star system made the idea simple, but it still connected back to the real meaning of weights in AI. The **bias** section was another part that helped make the concept easier. We showed the wizard with two stars and then added **one bonus star** as bias. That extra star changed the total from two stars to three stars, which allowed the wizard to attack. This helped us understand bias as an **extra push** that can affect the final result. The example was simple, but it showed an important idea: sometimes the AI has information from the inputs, but bias can still shift the result before the final decision is made.

The **activation function** was explained as the final rule. Our rule was simple: **three stars or more means ATTACK, and two stars or less means WAIT**. This made activation easier to understand because we could clearly see that something still had to take the information from the weights and bias and turn it into an actual choice. In our example, the wizard had two stars, received the bonus bias star, reached three stars, and then the activation rule chose ATTACK. Seeing all three ideas work together helped connect the whole process instead of treating weights, bias, and activation as separate topics. The interactive game was also an important part of the project because it let the audience follow the decision step by step. The process was basically **look at the clue, count the stars, and choose Attack or Wait**. That made the presentation more hands-on and also showed that the goal was not just to explain the words, but to make the audience understand the process. The project was designed for an 11-year-old audience, so keeping the explanations simple was important. We learned that making something simple does not mean leaving out the meaning. It can actually force us to understand the topic better because we have to explain it without hiding behind complicated vocabulary.

Another useful part of the presentation was connecting the wizard game to **real AI examples**. We showed that a car AI can look at the road and decide whether to stop or go, a voice AI can listen to words and decide how to answer, and a game AI can watch a player and decide whether to attack or hide. Those examples showed that the same basic idea appears in many different AI systems: look at information, decide what matters, and make a choice. Our wizard just gave us a simpler way to see that process. Overall, this project helped our group understand how **weights, bias, and activation functions work together**. The biggest takeaway was that AI decisions are built from steps. The AI looks at clues, gives more importance to some information, adds bias, and then follows a final rule to make a decision. The wizard and alien example made those steps easier to remember because everything connected to one story. Instead of seeing these terms as separate definitions, we now have a clearer picture of how they work together inside an AI decision.